

StockLite

Final Assignment - Intro to DevOps

Sinh vien	Nguyen Xuan Linh
San pham	StockLite - quan ly kho nho
Website	https://stocklite.404hz.me
Monitoring	https://grafana.404hz.me
Repository	https://github.com/NXL16/Stock_Lite

Muc tieu cua bao cao

Trình bày việc xây dựng và vận hành StockLite như một hệ thống thực tế: full-stack CRUD, deploy HTTPS, CI/CD tự động, monitoring và alert Telegram.

1. San pham va kien truc

StockLite la web app quan ly san pham va ton kho. Ngươi dung co the tao, xem, sua, xoa san pham; nhap/xuat kho va theo doi danh sach sap het hang. Du lieu duoc luu trong PostgreSQL nen van ton tai sau restart app va reboot VPS.

Kien truc

Internet -> Nginx HTTPS -> React static build va /api -> Express API (127.0.0.1:5002) -> PostgreSQL (127.0.0.1:5432). PM2 quan ly tien trinh stocklite-api. Cac cong noi bo khong expose truc tiep ra Internet.

API va giao dien

API cung cap day du GET list, GET detail, POST, PUT va DELETE cho entity Product. UI React co validate form, loading, xu ly loi, xac nhan xoa va thao tac nhap/xuat kho. Endpoint /api/health duoc dung trong deploy pipeline va monitoring.

2. Deploy va bao mat

Nginx phuc vu frontend va reverse proxy /api ve Express. HTTP redirect sang HTTPS voi chung chi Lets Encrypt/Certbot. PM2 tu restart khi crash va cau hinh startup giup app chay lai sau reboot.

PostgreSQL dung user va password riêng cho StockLite, chi nghe localhost. UFW chi cho phep SSH, 80 va 443. Secrets VPS, database va Telegram duoc luu ngoai Git; repo ignore .env va file Alertmanager that.

3. CI/CD va Telegram

Workflow CI chay tren moi push/pull request vao main. CI cai dependencies, chay unit test that cho logic ton kho va build React. Workflow CD chi duoc kich hoat khi CI thanh cong; CD pull source tren VPS, build frontend, startOrReload PM2, reload Nginx va health check /api/health.

Khi CI/test that bai, CD khong chay va production giu nguyen ban cu. Telegram gui mot tin bao ket qua pipeline: thanh cong kem nguoi push va commit message; that bai kem link log GitHub Actions.

4. Monitoring va alert

Prometheus thu thap CPU, RAM, disk tu Node Exporter va metrics HTTP tu endpoint /metrics cua Express. Grafana duoc truy cap qua HTTPS sau Nginx. Hai query da dung de quan sat app:

```
sum(rate(http_request_duration_seconds_count{job="stocklite-api"}[5m]))  
  
histogram_quantile(0.95, sum by (le)  
(rate(http_request_duration_seconds_bucket{job="stocklite-api"}[5m])))
```

Alertmanager co ba quy tac: StockLiteAPIDown sau 1 phut, HighCPU lon hon 80% trong 5 phut va DiskAlmostFull khi con duoi 15% trong 5 phut. Moi alert deu co huong dan xu ly va send_resolved.

5. Dien tap su co va ket luan

Dien tap su co duoc thuc hien bang pm2 stop stocklite-api. Sau khi Prometheus thay target down qua nguong for, Alertmanager gui Telegram FIRING. Sau pm2 start stocklite-api, target phuc hoi va Telegram gui RESOLVED. Dashboard Node Exporter va Grafana Explore the hien CPU/RAM/disk, request per second va p95 latency.

Bang chung can nop

Muc	Bang chung
Source va workflow	Repository public co backend, frontend, monitoring va .github/workflows
Website	StockLite chay HTTPS va thao tac CRUD
CI/CD	GitHub Actions xanh, deploy va Telegram notification
Monitoring	Node Exporter Full, request/s va p95 latency
Alert	Telegram FIRING va RESOLVED khi stop/start PM2
Video	CRUD -> deploy xanh -> test fail -> fire drill

Ket luan

StockLite dap ung nam tru cot cua Final Assignment: full-stack CRUD, deploy VPS HTTPS, CI/CD va notify, monitoring va alert co dien tap su co. README cung cap kien truc, cau hinh, runbook va kich ban demo de he thong co the duoc dung lai va van hanh nhat quan.